

Assignment - 1

NAME - Akshay sharma

ROLL NO. - 25EARIT009

Question 1

What is a variable in C/C++?

Ans: variable is a named storage location in the computer's memory used to hold data that a program can manipulate

Question 2

- Why do we use variables in programs?
- Ans: variables are used as named containers to temporarily store, manipulate, and retrieve data within a computer's memory during a program's execution

Question 3

- What is the difference between a variable name and its value?
- Ans: The variable name is the identifier, while the value is the actual data stored in it.

Question 4

- Which of the following is a valid variable name? a) 2num b) total_marks c) for d) my marks
- Ans: b) total_marks

Question 5

- Why is Age different from age?
- Ans: C++ is case-sensitive, so Age and age are different variables

Question 6

- Can a variable name contain spaces? Why?
- Ans: No. Variable names cannot contain spaces.

Question 7

- Can a variable name start with an underscore (`_`) ?
- Ans: Yes , variable names can start with an underscore . example: `_count`

Question 8

- What happens if two variables have the same name in the same scope?
- Ans: The compiler generates an error because variable names must be unique within the same
- scope

Question 9

- Write three meaningful variable names for storing student information.
- ANS : studentName, studentAge, studentMarks

Question 10

- Which naming convention is better and why: studentMarks or sm?
- Ans: studentMarks because it is descriptive and easy to understand

Question 11

- What is a data type?
- Ans A data type specifies the kind of data a variable can store.

Question 12

- Which data type is used to store whole numbers?
- Ans : int

Question 13

- Which data type is used to store decimal values?
- Ans: float or double

Question 14

- What is the difference between float and double?
- Ans: double provides greater precision than float.

Question 15

- Which data type stores a single character?
- Ans: char

Question 16

- Which data type stores True/False values?
- Ans: bool

Question 17

- What data type is suitable for storing a person's name?
- Ans: string

Question 18

- Predict the output: `int age=18; cout<<age;`
- Ans: 18

Question 19

- Identify the data type: 25, 3.14, 'A', true.
- Ans: 25=int, 3.14=double, 'A'=char, true=bool

Question 20

- What happens if you store a decimal number in an int variable?
- Ans: The decimal part is discarded. Example:
3.14 becomes 3

Question 21

- What is the purpose of cin?
- Ans: cin is used to take input from the user

Question 22

- What is the purpose of cout?
- Ans: cout is used to display output on the screen.

Question 23

- Which symbol is used with cin?
- Ans: >>

Question 24

- Which symbol is used with cout?
- Ans: <<

Question 25

- Explain: `cin >> age;`
- Ans: It takes input from the user and stores it in age.

Question 26

- Explain: `cout << age;`
- Ans: It displays the value of age

Question 27

- What is the purpose of endl?
- Ans: endl moves the cursor to the next line.

Question 28

- Write a statement to input a student's marks.
- Ans: `cin >> marks;`

Question 29

- Write a statement to display 'Welcome to C++'.
- Ans: `cout << "Welcome to C++";`

Question 30

- Predict the output: `cout << 'Programming';`
- Ans: Error because single quotes are for a single character.

Question 31

- What is an operator?
- Ans: An operator is a symbol that performs an operation on operands

Question 32

- Which operator is used for addition?
- Ans:+

Question 33

- What is the result of $10 \% 3$?
- Ans: 1

Explanation: The $\%$ (modulus)operator gives the remainder after division.

$$10 \% 3 = 1$$

Because: $10/3=3$ remainder 1.

Question 34

- What is the result of $8 \% 2$?
- Ans:0
- Because:8 is completely divisible by 2,so remainder is 0.

Question 35

- Which operator checks equality?
- Ans:==

Question 36

- What is the difference between = and == ?
- Ans:= assigns a value, while == compares two values.

Question 37

- What is the result of $5 > 3$?
- Ans:True

Question 38

- What does && represent?
- Ans: Logical AND operator.It returns true only when both conditions are true.

Question 39

- What does `||` represent?
- Ans: Logical OR operator. It returns true if at least one condition is true .

Question 40

- What does ! represent?
- Ans: Logical NOT operator.It reverses the result of a condition.

Question 41

- Write a program to print "Hello world"?

- **Problem Statement:**

- Write a program to display "Hello world " on screen.

- **Algorithm:** 1. start the program.
2. Display "Hello world" using cout
3. End the program.

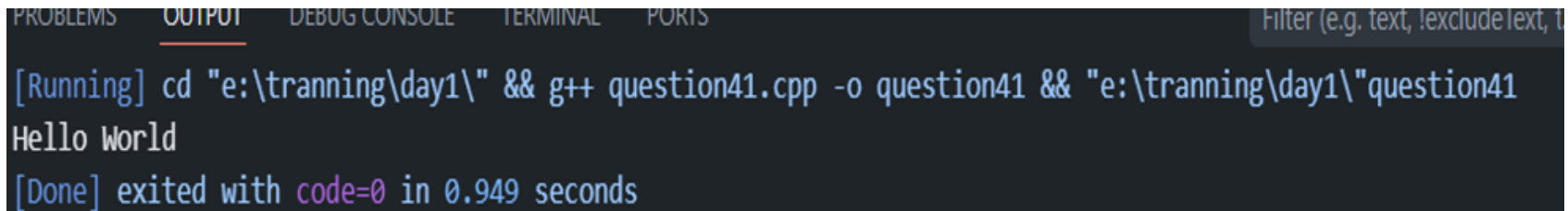
- **C++ Program**

```
#include<iostream>
using namespace std;
int main()
{
    cout<<"Hello world";
    return 0;
}
```

- **Explanation :**

- `#include <iostream>`: used for input/output operations
- `using namespace std;` → Allows use of cout without `std::`

- `cout` → Displays output on screen
- `return 0;` → Ends the program successfully
- **Output**



The screenshot shows an IDE's output window with tabs for PROBLEMS, OUTPUT, DEBUG CONSOLE, TERMINAL, and PORTS. The OUTPUT tab is selected. The text in the output window is as follows:

```
[Running] cd "e:\tranning\day1\" && g++ question41.cpp -o question41 && "e:\tranning\day1\"question41
Hello World
[Done] exited with code=0 in 0.949 seconds
```

A search filter bar at the top right of the output window contains the text "Filter (e.g. text, !exclude text, t".

Question 42

- Write a program to input two numbers and display their sum.
- **Problem Statement**
- Write a program to take two numbers from the user and display their sum.
- **Algorithm**
 1. Start the program
 2. Declare two variables
 3. Take input from user
 4. Add both numbers
 5. Display the result
 6. End the program
- **C++ Program**

```
#include<iostream>
Using namespace std;
int main()
{
int num1,num2,sum;
```

```
cout<<"enter first number:";  
cin>>num1;  
sum=num1 +num2;  
cout<<"sum="<<sum;  
Return 0;  
}
```

Explanation

```
day1> g++ question42.cpp -o question42
```

```
/lib/gcc/mingw32/6.3.0/../../../../mingw32/bin/ld.exe: cannot open output file question42.exe: P  
rror: ld returned 1 exit status
```

```
ber: 10
```

```
mber: 20
```

Question 43

- Write a program to calculate the area of a rectangle.

- **Problem Statement**

Write a program to calculate the area of a rectangle.

- **Formula:**

Area = Length $\times$ Breadth

- **Algorithm**

1. Start the program
 2. Declare variables for length and breadth
 3. Take input from user
 4. Calculate area
 5. Display area
1. End the program

- **C Program**

```
#include<stdio.h>
int main()
{
```

```
float length, breadth, area;  
    printf("Enter length: ");  
    scanf("%f", &length);  
    printf("Enter breadth: ");  
    scanf("%f", &breadth);  
    area = length * breadth;  
    printf("Area of Rectangle = %.2f", area);  
    return 0;  
}
```

Explanation

- scanf() takes input
- length * breadth calculates area
- printf() displays output
- %.2f shows decimal values up to 2 places

```
• >> ./question43  
Enter length: 2  
Enter breadth: 6
```

Question 44

- What formula is used to calculate Simple Interest?
- **Ans:**The formula used to calculate Simple Interest (SI) is:

$$SI = P * R * T / 100$$

Where:

- P = Principal Amount
- R = Rate of Interest
- T = Time
- SI = Simple Interest

Write a program to calculate Simple Interest.

- **Problem Statement**

Write a program to calculate Simple Interest using principal, rate, and time.

- **Algorithm**

1. Start the program
 2. Declare variables
 3. Take principal, rate, and time as input
 4. Apply formula
 5. Display result
1. End the program

C Program

```
#include<stdio.h>
int main()
{
Float principal,rate,time,Sl;
printf("enter principal amont:");
scanf("%f",&principal);
```



```
printf("Enter Rate of Interest: ");
scanf("%f", &rate);
printf("Enter Time: ");
scanf("%f", &time);
SI = (principal * rate * time) / 100;
printf("Simple Interest = %.2f", SI);
return 0;
}
```

Explanation

- User enters principal amount, rate, and time
- Formula is applied to calculate simple interest
- Result is displayed using printf()

```
PS E:\training\day1> gcc question45.c -o question45
• >> ./question45
Enter Principal Amount: 10000
Enter Rate of Interest: 7
Enter Time: 3
Simple Interest = 2100.00
```

Question 46

- Why is a temporary variable used while swapping two numbers?
- **Ans :** A temporary variable is used to safely store one value while swapping two numbers so that the original value is not lost.

QUESTION 47

Swap the values: a=10 b=20

- **Problem Statement**

Write a program to swap two numbers using a temporary variable.

- **Algorithm**

1. Start the program
 2. Declare variables a, b, and temp
 3. Store value of a in temp
 4. Assign value of b to a
 5. Assign value of temp to b
 6. Display swapped values
1. End the program

C++ Program

```
#include<iostream.h>
```

```
Using namespace std;
```

```
int main()
```

```
{
```

```
int a=10,b=20,temp;
```

```
cout << "Before Swapping:" << endl;
    cout << "a = " << a << endl;
    cout << "b = " << b << endl;
    temp = a;
    a = b;
    b = temp;
    cout << "After Swapping:" << endl;
    cout << "a = " << a << endl;
    cout << "b = " << b << endl;
    return 0;
}
```

Explanation

- temp = a stores value of a
- a = b copies value of b into a
- b = temp restores original value of a into b
- This swaps both numbers successfully

```
>> ./question4/
Before Swapping:
a = 10
b = 20
After Swapping:
a = 20
```

Question 48

- What formula is used to convert Celsius into Fahrenheit?
- **Ans:** $F = \frac{9}{5}C + 32$
- **Explanation:**
C= temperature in Celsius
F= temperature in fahrenheit

Question 49

- What is the Fahrenheit value of 0°C ?
- **Ans:** using the formula $F = \frac{9}{5}C + 32$
Where $C = 0^{\circ}\text{C}$
- **Calculation:**
$$F = \left(\frac{9}{5} \times 0\right) + 32$$
$$F = 32^{\circ}\text{F}$$
- **Final Answer:**
The Fahrenheit value of 0°C is 32°F .

Why can integer division give incorrect results in the Celsius-to-Fahrenheit program?

Answer

Integer division can give incorrect results because integers remove the decimal part.

For example:

5 / 9

This becomes:

0

instead of:

0.555...

So, if integer division is used in the Celsius-to-Fahrenheit formula, the answer may become wrong.

Correct Way

Use float or double for accurate decimal calculations.